CLASS 8 · CBSE · GANITA PRAKASH PART 2 · SHORT NOTES

Chapter 7: Area

Topic-wise short notes with solved examples

Area as a count of unit squares, and the formulas for the area of a rectangle, triangle, parallelogram, rhombus and trapezium, obtained by cutting and rearranging shapes. Also areas of paths, areas of polygons and units of area.

1 Area and perimeter

- The **area** of a region is the number of unit squares (including parts of squares) needed to cover it exactly. Units: sq cm (cm^2), sq m (m^2), and so on.
- **Area of a rectangle** = length × width. **Area of a square** = side × side.
- **Perimeter does not measure area.** Two regions can have the same perimeter and different areas, and a region with a larger perimeter can have a smaller area.

Rectangle	Perimeter	Area
7 cm by 4 cm	22 cm	28 sq cm
8 cm by 3 cm	22 cm	24 sq cm
10 cm by 2 cm	24 cm	20 sq cm

- If every length of a figure is doubled, its area becomes **4 times**.

Example 1.1 A rectangle of area 35 sq in has one side 7 in. Find the other side.

SOLUTION

$35 \div 7$

5 in

Example 1.2 Give two rectangles where the one with the larger perimeter has the smaller area.

SOLUTION

10 by 1 (perimeter 22, area 10) and 5 by 5 (perimeter 20, area 25)

2 Paths

- **Path around a rectangle**: area of the path = area of the outer rectangle − area of the inner rectangle. Moving the inner rectangle about inside the outer one does not change this.
- A path of width w **outside** a park l by b has area $(l + 2w)(b + 2w) - lb$.
- **Cross-paths** of widths w_1 and w_2 running across a plot l by b, one along the length and one along the breadth: area $= lw_1 + bw_2 - w_1w_2$. The small rectangle where they cross must not be counted

twice.

Example 2.1 A park 30 m by 20 m has a path 2 m wide all around it on the outside. Find the area of the path.

SOLUTION

Outer rectangle: 34 m by 24 m, area 816 sq m. Park: 600 sq m.

216 sq m

Example 2.2 A plot 14 m by 12 m has two cross-paths, each 2 m wide, one parallel to each side. Find the area of the paths.

SOLUTION

$14 \times 2 + 12 \times 2 - 2 \times 2$

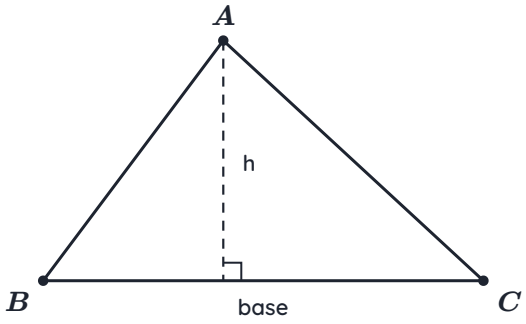
48 sq m

3 Triangle

- A diagonal divides a rectangle into two congruent triangles, so a right triangle is half of a rectangle.
- Any triangle is half of the rectangle that has the same base and the same height.

$$\text{Area of a triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

- The **height** (altitude) is the perpendicular distance from the opposite vertex to the line containing the base. For an obtuse-angled triangle it may fall outside the triangle; the formula still holds.
- Any side can be taken as the base, with its own height. So $\frac{1}{2} \times b_1 \times h_1 = \frac{1}{2} \times b_2 \times h_2$.



Area = ½ × base × height

Example 3.1 Find the area of a triangle with base 5 cm and height 3.2 cm.

SOLUTION

$\frac{1}{2} \times 5 \times 3.2$

8 sq cm

Example 3.2 In triangle ABC, $BC = 6$ units and the altitude from A is 3 units. If $AC = 4$ units, find the altitude from B to AC.

SOLUTION

Area $= \frac{1}{2} \times 6 \times 3 = 9$. Also area $= \frac{1}{2} \times 4 \times h$.

$h = 4.5$ **units**

4 Useful facts about triangle areas

- A **median** (the line from a vertex to the mid-point of the opposite side) divides a triangle into two triangles of **equal area**: they have equal bases and the same height.
- The two diagonals of a rectangle (or any parallelogram) divide it into **four triangles of equal area**.
- **Triangles on the same base and between the same parallel lines have equal areas**, because they have the same height.
- Among all such triangles, the **isosceles** one has the smallest perimeter. (Reflect one base vertex in the parallel line; the shortest path is a straight line.)
- Joining the mid-points M and N of two sides of a triangle cuts off a triangle with $\frac{1}{4}$ of the area.

Example 4.1 The diagonals of a rectangle 8 cm by 6 cm meet at O. Find the area of each of the four triangles formed.

SOLUTION

The rectangle has area 48 sq cm and the four triangles are equal in area.

12 sq cm each

Example 4.2 Triangle SUB is isosceles with $SU = SB$, and SE is perpendicular to UB. If triangle SEB has area 24 sq units, find the area of triangle SUB.

SOLUTION

In an isosceles triangle the altitude to the base bisects the base, so the two halves are equal in area.

48 sq units

5 Any polygon

- Every polygon can be divided into triangles by drawing diagonals. Add the areas of the triangles.
- **Quadrilateral** with diagonal d and perpendiculars h_1, h_2 on it from the other two vertices: area $= \frac{1}{2} \times d \times (h_1 + h_2)$.

Example 5.1 In quadrilateral ABCD, diagonal $AC = 22$ cm and the perpendiculars from B and D on AC are 3 cm each. Find the area.

SOLUTION

$\frac{1}{2} \times 22 \times (3 + 3)$

66 sq cm

Example 5.2 A regular hexagon has side 4 cm. Find its area. (An equilateral triangle of side s has area $\frac{\sqrt{3}}{4}s^2$.)

SOLUTION

It is made of 6 equilateral triangles of side 4 cm, each of area $4\sqrt{3}$ sq cm.

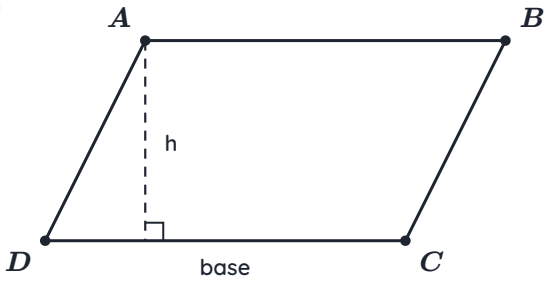
$24\sqrt{3}$ sq cm, about 41.6 sq cm

6 Parallelogram

- Cut a right triangle off one end of a parallelogram and move it to the other end: the result is a rectangle with the same base and height. Cutting and rearranging like this is called **dissection**.

$$\text{Area of a parallelogram} = \text{base} \times \text{height}$$

- The height is the perpendicular distance between the base and the opposite side, **not** the slant side.
- Either pair of sides can be used: $b_1 \times h_1 = b_2 \times h_2$.
- Parallelograms on the same base and between the same parallels have equal areas but different perimeters. A rectangle has a greater area than a slanted parallelogram with the same side lengths.



Area = base × height

Example 6.1 Find the area of a parallelogram with base 7 cm and height 4 cm.

SOLUTION

28 sq cm

Example 6.2 In parallelogram PQRS, $SR = 12$ cm and the height on SR is 7.6 cm. If $QR = 8$ cm, find the height on QR.

SOLUTION

Area $= 12 \times 7.6 = 91.2$ sq cm. Then $8 \times h = 91.2$.

11.4 cm

7 Rhombus

- A rhombus is a parallelogram, so area = base × height.
- Its diagonals are perpendicular bisectors of each other, which gives a second formula:

$$\text{Area of a rhombus} = \frac{1}{2} \times d_1 \times d_2$$

- Reason: the diagonal d_2 cuts the rhombus into two triangles, each with base d_2 and height $\frac{d_1}{2}$.
- The same formula works for a kite and for any quadrilateral with perpendicular diagonals.

Example 7.1 Find the area of a rhombus with diagonals 20 cm and 15 cm.

SOLUTION

$\frac{1}{2} \times 20 \times 15$

150 sq cm

Example 7.2 A rhombus has area 96 sq cm and one diagonal 12 cm. Find the other diagonal and the side.

SOLUTION

$\frac{1}{2} \times 12 \times d = 96$ gives $d = 16$ cm. Half-diagonals 6 and 8 give side $\sqrt{36 + 64}$.

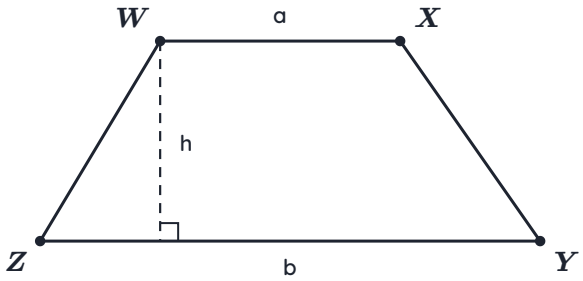
16 cm; side 10 cm

8 Trapezium

- For a trapezium with parallel sides a and b and height h (the perpendicular distance between them):

$$\text{Area of a trapezium} = \frac{1}{2} \times h \times (a + b)$$

- **Why**: two copies of the trapezium, one turned upside down, fit together to make a parallelogram with base $a + b$ and height h. The trapezium is half of it.
- It can also be found by splitting the trapezium into a rectangle and two right triangles.



Area = ½ × h × (a + b)

Example 8.1 The parallel sides of a trapezium are 24 m and 36 m, and the distance between them is 14 m. Find its area.

SOLUTION

$\frac{1}{2} \times 14 \times (24 + 36)$

420 sq m

Example 8.2 A trapezium has area 144 sq cm and parallel sides 10 cm and 14 cm. Find its height.

SOLUTION

$\frac{1}{2} \times h \times 24 = 144$

12 cm

Example 8.3 A regular hexagon is divided into a trapezium (half the hexagon), a rhombus and an equilateral triangle. Find the ratio of their areas.

SOLUTION

The hexagon is 6 small equilateral triangles: the trapezium uses 3, the rhombus 2 and the triangle 1.

$3 : 2 : 1$

9 Changing shape, keeping area

- The Śulba-Sūtras describe how to turn one shape into another of the same area, as needed for building altars.

From	To	Method
Triangle	Rectangle	Join the mid-points of two sides; the rectangle has the same base and half the height
Isosceles triangle	Rectangle	Cut along the altitude and join the two halves along their longest sides
Parallelogram	Rectangle	Cut off a right triangle at one end and move it to the other
Rhombus	Rectangle	Sides d_1 and $\frac{d_2}{2}$
Trapezium	Rectangle	Cut through the mid-points of the slant sides; length $\frac{a+b}{2}$, height h

- A rectangle with **twice** the area of a triangle: use the same base and the same height.

10 Units of area

Conversion	Value
1 inch	2.54 cm
1 foot	12 inches
1 sq inch	$2.54^2 = 6.4516$ sq cm
1 sq foot	144 sq inches
1 sq m	10,000 sq cm
1 sq km	10,00,000 sq m
1 acre	43,560 sq ft
1 hectare	10,000 sq m

- To convert an **area**, use the **square** of the length conversion factor.
- An A4 sheet is 21 cm by 29.7 cm: area 623.7 sq cm.

Example 10.1 Convert 10 sq in to sq cm, and 161.29 sq cm to sq in.

SOLUTION

$10 \times 6.4516 = 64.516$ and $161.29 \div 6.4516 = 25$

64.516 sq cm; 25 sq in

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11 Common mistakes

- Using a slant side as the height of a parallelogram, triangle or trapezium.
- Forgetting the $\frac{1}{2}$ in the triangle, rhombus and trapezium formulas.
- Multiplying the sides of a rhombus instead of using the diagonals.
- Counting the crossing square twice in cross-path problems.
- Converting area with the length factor: 1 sq ft is 144 sq in, not 12.
- Assuming that equal perimeters mean equal areas.

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